Compose, abort error, 124
Compose, already running error, 50–51
connecting to database, 79
inotify overflow, 90
Rails server, 50–51, 110, 121–123
shutting down containers, 49
eval, 158
exec, 55, 74, 99, 123
Exec form of CMD, 33
execute permissions, 123
exit, 12

## F

-f option, 54, 83, 151
FaaS, serverless computing, 140–141
Fargate, 141
Fig, 60
file ownership
connecting Redis to app, 66
modifying files, 13
platform differences, 199
testing, 97, 100, 104
using databases, 80
files, *see also* file ownership; mounting
adding to containers, 21
config file structure, 145
copying, 21, 40
directory for project files, 11
editing locally, 13
filename option, 151
ignoring unnecessary, 34, 37
names, 78
precompiling assets, 92
separating files that trigger rebuilds, 40
sharing config between Compose files, 195
follow output option, logs, 54
--force-recreate option, 79
Foreman, 67
FROM, 19, 24, 194
Function as a Service (FaaS), serverless computing, 140–141

## G

gem management, 65, 113–119
Gemfile, 22, 38–41
gems
caching, 38–41
installing, 22, 117
GKE, *see* Google Kubernetes Engine
Google Cloud Container Registry, 153
Google Compute Engine, xv, 135
Google Kubernetes Engine, 138, 140

## H

-h option, 64
HAProxy, 192
Harbor, 153
headless testing, 102, 108
Hello World example, 92
holistic view, xiii–xiv
host connecting option, 64
host network, 63
hostnames
DNS resolution, 64
JavaScript testing, 104
load balancing with Swarm, 171
Registries, 150
services, 64
HTTPS, 89
Hyper-V, 5, 156

## I

-i option, 11, 13
IaaS
vs. CaaS, 142
platforms, 134–135
IDs
Docker ID, 149
finding IP addresses of running containers, 29
generated, xvii
throwaway containers, 24
if, 123
image property, 61
images
about, xiv
automatic downloading, 8, 24
base, 20
build cache, 36–41
building, 23–25
building with Compose, 46–47, 55
building, multi-stage, 194
caching layers, 9
copying, 147
dangling, 56
defined, 6
defining custom, 18–23
finding to use, 201–203
getting into new instances, 159
intermediate, 24
JavaScript front end, 88
labels, 42
listing, 25
names, 25, 31–33, 48, 150
Official Images, 151
precompiling assets for production, 147
pruning, 56
pushing, 151–154
rebuilding with Compose, 55
registries, 149–154
running a Rails server with, 25
running specific version of, 33
security, 192, 202
sharing, 148–154
specifying for services, 61
stack file, 162
starting, 19
understanding run, 8–10
versioning, 32, 48
images command, 25
Infrastructure as a Service, *see* IaaS
infrastructure as code, 18
inotify overflow error, 90
input options, 13, 15
inspect, 29, 84
install command, 11
installation
with apt get, 20
caching gems, 38–41
Capybara, 99
Compose, 4
Docker, 3–6
Docker advantages with, xiv
gems, 22, 117
Node.js, 20, 88
packages and caching issue, 37